CS & IT ENGINEERING

COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 1

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Recap of Previous Lecture

doubt ?

Topic Multiple Chips in Single Memory System

Topic DRAM Refresh

Topic Associative Memory

Topics to be Covered

Topic Associative Memory

Topic Locality of Reference

Topic Cache Memory

[NAT]

59 to 60

#Q. A 32–bit wide main memory unit with a capacity of 1 GB is built using 256M X 4-bit DRAM chips. The number of rows of memory cells in the DRAM chip is 2^{14}. The time taken to perform one refresh operation is 50 nanoseconds. The refresh period is 2 milliseconds. The percentage (rounded to the closet integer) of the time available for performing the memory read/write operations in the main memory unit is ______?

chip refresh time = $2^{14} * 50$ ns

$2^{10} = 1024$

$2^{10} = 1K$

$= 819200$ ns

$= 0.8192$ ms

$= 800$ μsec

$= 0.8$ ms

% of time remaining for R/W

$$= \frac{2 - 0.8192}{2} * 100\%$$

$$= 59\%$$

$$= \frac{2 - 0.8}{2} * 100\%$$

$$= 60\%$$

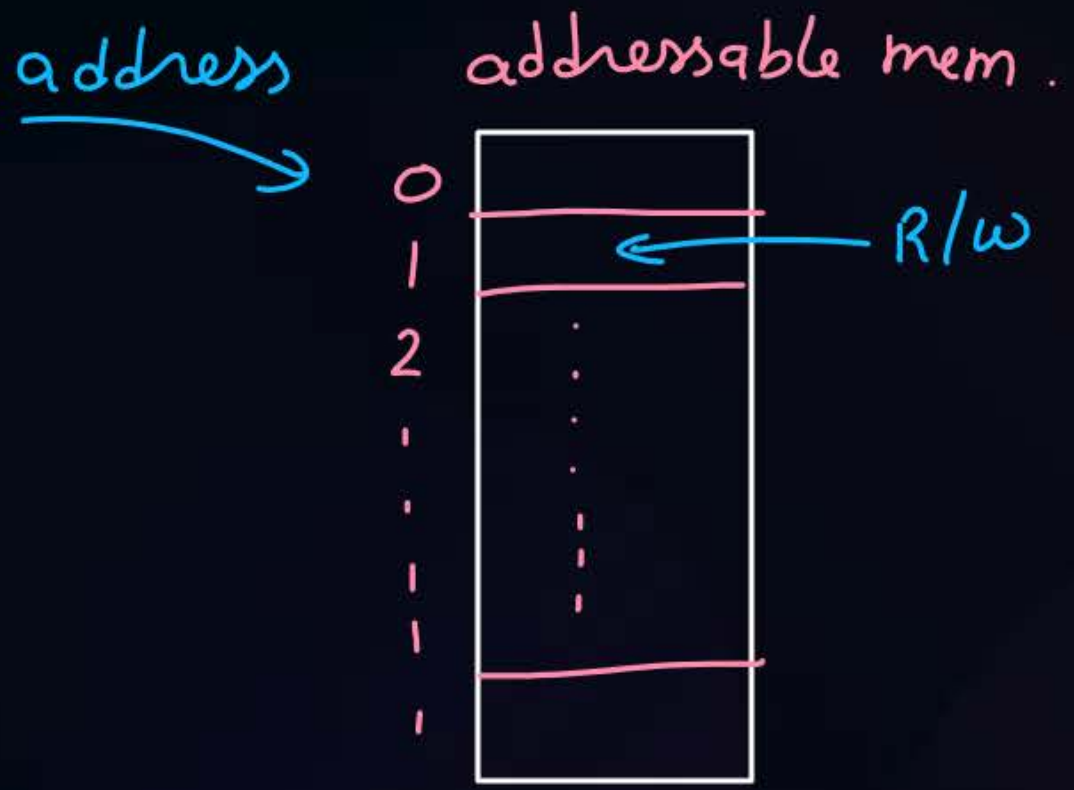
Topic : Associative Memory

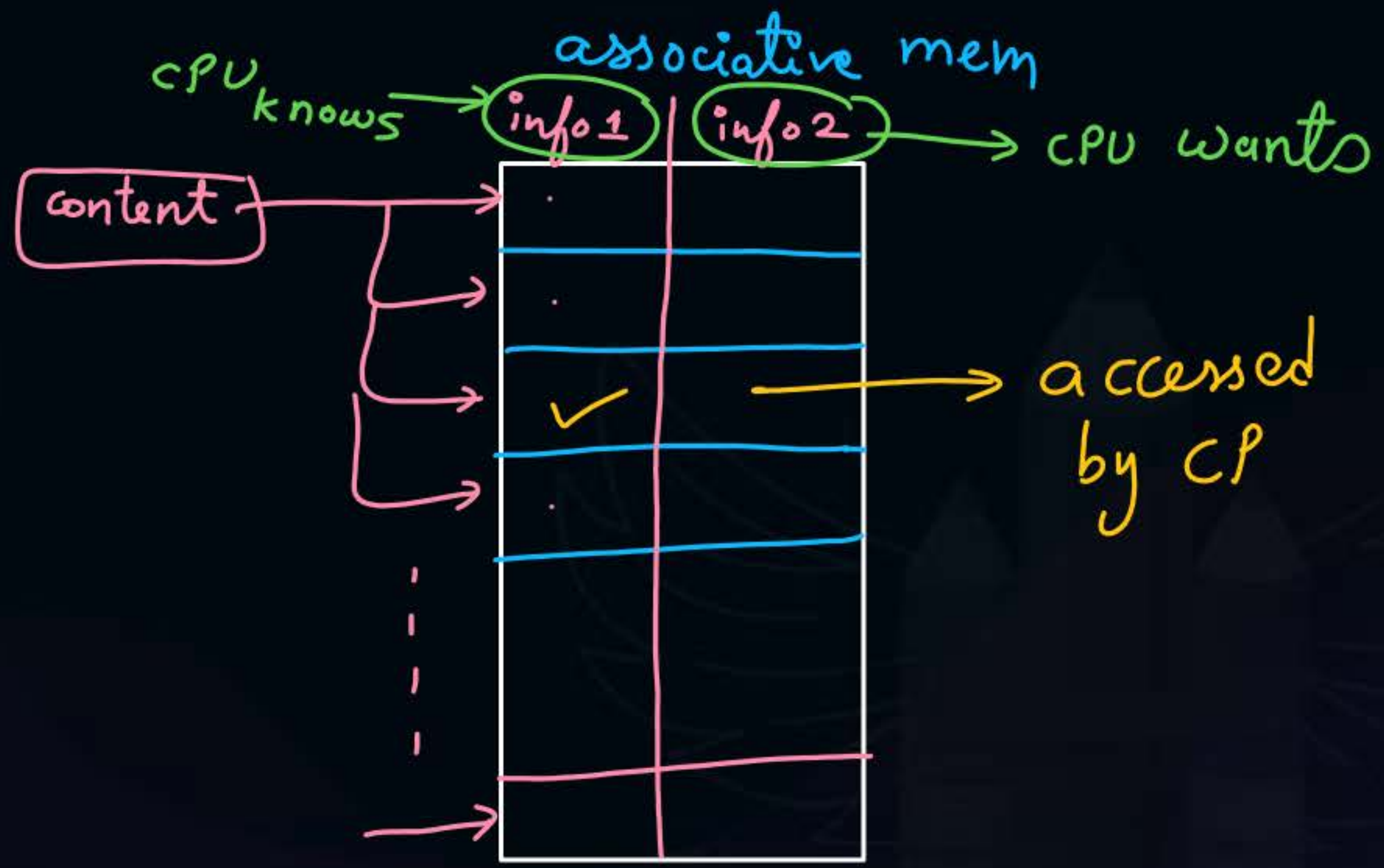
CAM

Known as content addressable memory also

→ very expensive & very expensive

⇒ cells do not addresses





→ used for implementing cache & TLB (Translation Lookaside Buffer)

Topic : Locality of Reference

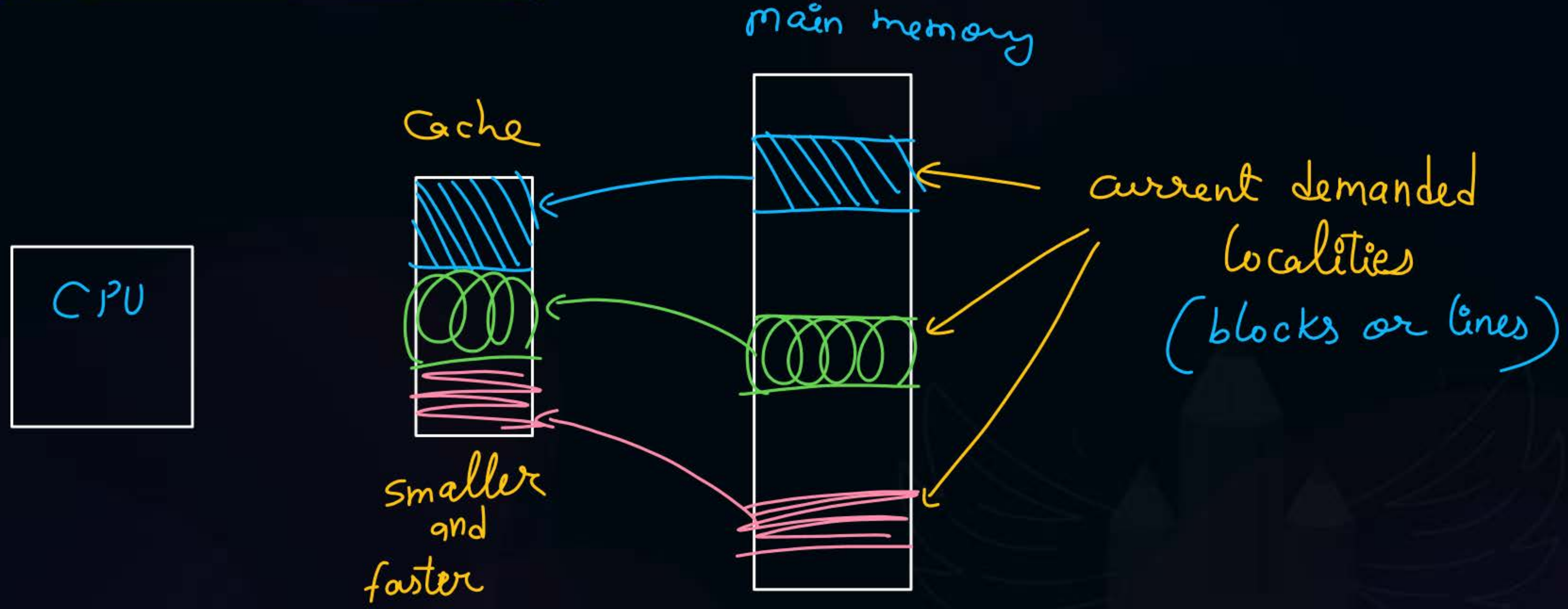
If CPU has requested one address for memory access, then that particular address or near by addresses will be accessed soon.

Topic : Locality of Reference

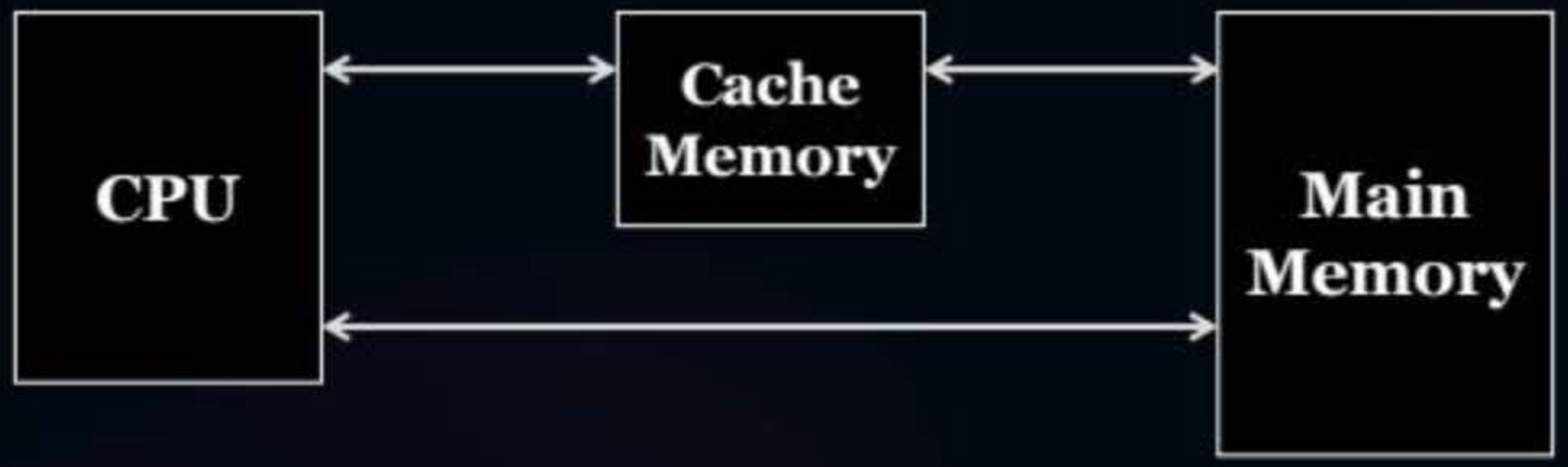
Types:

1. Spatial (according to space) ⇒ if nearby addresses are accessed soon
2. Temporal (according to time) ⇒ if same address is — " —

Topic : Cache Memory



Topic : Cache Memory



Use of cache reduces CPU's mem. access time.

Topic : Working of Cache Memory

1. Cache Hit ⇒ if CPU's demanded content is present in Cache.
2. Cache Miss ⇒ — " — is not — " —
3. Hit Ratio (H or h) :-

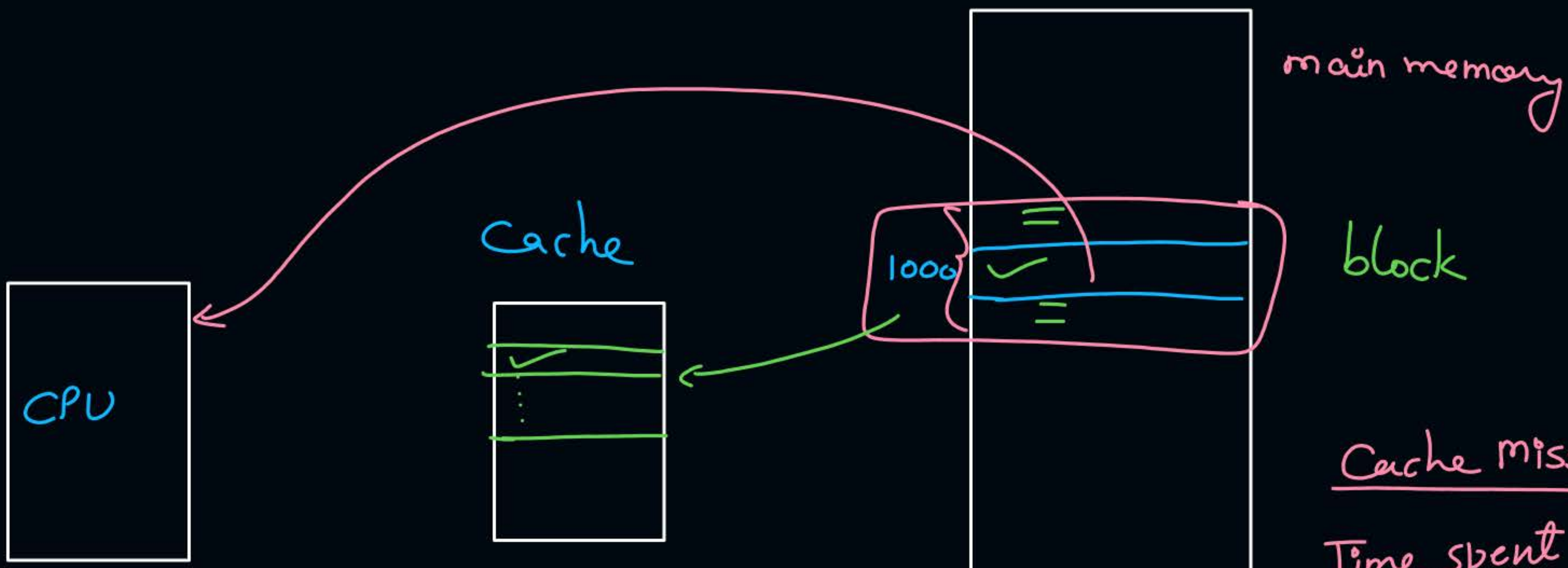
$$\text{hit ratio} = \frac{\text{no of hits in memory}}{\text{Total mem. references}}$$

$$\text{miss ratio} = 1 - h$$

$$= 100\% - h\%$$

example

$$= \frac{36}{40} = 0.9 \text{ or } 90\%$$



Cache miss Penalty :-

Time spent by CPU to bring missed block from mm to cache.

when there is a cache miss CPU reads demanded content from mm. But the block which contain demanded missed content is copied into cache.

Topic : Average Memory Access Time

(T_{avg})

$T_{avg} = H * $ Time required to access memory for cache hit $+ (1-H) *$ Time required to access memory for cache miss

.... 1

Topic : Types of Cache Accesses

1. **Simultaneous Access:** (Parallel)

Request for cache and main-memory are generated simultaneously

$$T_{avg} = H * t_{cm} + (1-H)\, t_{mm} \quad \ldots\ldots (2)$$

t_{cm} = cache access time

t_{mm} = mm —||—

Topic : Types of Cache Accesses

2. **Hierarchical Access:** (Serial)

Only cache is accessed first

$$T_{avg} = H * t_{cm} + (1-H)(t_{cm} + t_{mm})$$

$$= \cancel{H t_{cm}} + t_{cm} - \cancel{H t_{cm}} + (1-H) t_{mm}$$

$$= t_{cm} + (1-H) t_{mm} \quad \ldots\ldots (3)$$

Topic : When to Use Which Formula

if direct info^n of
hit time and miss time given

yes

Use general
formula
eq^n 1

no

$t_{cm}, t_{mm} \Rightarrow$ given

if hierarchy or level given

No

use simultaneous
eq^n 2

[cache search time is ignored or negligible]

[Cache lookup —— // ——————]

yes

use hierarchical
eq^n 3

Default

if PSUs or govt. exam ⇒ simultaneous

if GATE ⇒ any of simultaneous or hierarchical

if Test series of other institutes ⇒ Hierarchical

2 mins Summary

Topic Associative Memory

Topic Locality of Reference

Topic Cache Memory

Happy Learning

THANK - YOU